RESEARCH ARTICLE

Cozie Singapore: Scalable crowdsourced smartwatch micro-surveys to capture in-situ urban heat and noise perception

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ABSTRACT
The everyday experience of urban space is shaped as much by how people perceive heat, noise, and their social surroundings as by the physical conditions that sensors record. This paper characterizes how that lived experience differs across the common spaces of a dense tropical city using crowdsourced, watch-based micro-surveys collected with the open-source Cozie smartwatch platform. A field deployment in Singapore with 106 participants produced 9,962 geolocated micro-survey responses, each pairing self-reported thermal preference, noise, social context, and activity with the wearer's onboard physiological signals and with linked outdoor weather. Every response included a self-reported location in one of six everyday space types, and the reported experience was characterized through conditional perception fingerprints, relative distinctiveness expressed as lift, and effect-size-based association testing using the chi-square statistic, Cramér's V, and adjusted standardized residuals. The strongest separations between spaces are behavioral and acoustic rather than thermal, as what a person is doing and hearing identifies a space more sharply (Cramér's V up to 0.38) than how warm they feel (0.12). Commute spaces carry the loudest and most traffic-driven noise, indoor spaces are defined by talking, and cooling demand is lowest in air-conditioned offices and highest outdoors, while thermal preference tracks the immediate controlled environment more than the outdoor weather. These findings show that a scalable smartwatch micro-survey can recover and quantify the perceptual signature of everyday urban spaces, offering a descriptive, human-centered baseline for targeting noise mitigation and energy-conscious, thermal comfort-aware urban design in tropical cities.

KEYWORDS
Micro-ecological momentary assessments; Wearable sensing; Thermal preference; Urban noise perception; Human-centered urban design; Tropical cities

1. Introduction

A key component of urban studies is the characterization of how the built environment impacts humans. Often, this analysis uses conventional geospatial information from the data-rich mapping context. These geospatial contexts can then be enriched with a significant amount of metadata through convergence with temporal sensor data, imaging data from satellites or street-view imagery, or from smart devices. Street-view

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imagery has been used to quantify visual and perceived qualities of the streetscape at scale (Ito, Kang, Zhang, Zhang, & Biljecki, 2024), while network-based methods characterize the morphology and connectivity of the urban fabric (Yap, Stouffs, & Biljecki, 2023). Even human presence itself can be inferred indirectly, for example by estimating building occupancy from mobile phone traces (Barbour et al., 2019).

Despite this effort, significant elements of the human experience in the urban context are missing as observation is not the equivalent of capturing perception. Physical measurements alone are weak predictors of how occupants actually judge their surroundings, because non-physical and psychological factors shape perception (Castaldo et al., 2018), and expectations condition the same measured conditions differently (Schweiker, Rissetto, & Wagner, 2020). Even the most widely used comfort model predicts individual thermal sensation with limited accuracy across large field datasets (Cheung, Schiavon, Parkinson, Li, & Brager, 2019), underscoring the gap between what sensors record and how conditions are judged. Image-based surveys offer one way to elicit perception at scale, asking people to rate qualities such as safety, liveliness, and greenery from street-view imagery (Gu et al., 2025; Ito et al., 2024; Quintana et al., 2025). Despite this scalability, it is difficult for a person to judge the full experience of being in a space from an image alone. In-situ methods such as ecological momentary assessment and experience sampling have long provided a means of recording subjective states in real-world settings (Christensen, Barrett, Bliss-Moreau, Lebo, & Kaschub, 2003; Stone & Shiffman, 1994), and pairing these records with location data makes them geographically explicit (Kirchner & Shiffman, 2016).

Despite this methodological basis, in-situ perception data rarely span the full range of settings a person moves through in a day. As a result, it remains unclear how the experience of heat, noise, and social context differs between homes, workplaces, classrooms, transport, and outdoor spaces, particularly in dense tropical cities. This study addresses that gap with a city-scale deployment of watch-based micro-surveys in Singapore. Its objective is to characterize the self-reported experience of everyday urban spaces and the coverage of the linked wearable and weather data. Two research questions guide the analysis. The first asks how closely self-reported thermal preference and noise distraction track the linked outdoor heat and measured sound level, and how much they vary between participants. The second asks how thermal preference, noise, social context, and activity differ across six everyday space types, and which of these dimensions separate the spaces most strongly. The study is descriptive and associational by design. It characterizes what participants reported in each self-reported space type and uses the linked physiological and weather streams to contextualize thermal preference and noise. It does not estimate causal or mediated effects of specific environmental or design parameters. Section 2 reviews related work and prior analyses of the same deployment, and Section 3 describes the deployment, dataset, and analytical methods. Section 4 addresses the two research questions, and Sections 5 and 6 discuss implications, limitations, and future directions.

2. Related work

Three strands of prior work frame the present study. The first concerns methods for capturing perception in situ, the second concerns heat and noise as everyday exposures, and the third is earlier work with the Cozie platform.

2.1. Crowdsourcing humans-as-a-sensor data across the urban context

Geographically-explicit ecological momentary assessment (GEMA) has been pioneered as a means of combining subjective human information with its spatial context. It has been used in numerous applications, particularly in public health and research on place and well-being (Kirchner & Shiffman, 2016). GEMA has been used to relate noise annoyance to activity context and the daily acoustic environment (Zhang, Zhou, Kwan, Su, & Lu, 2020) and to compare mental well-being with exposure to natural diversity (Hammoud et al., 2024). These applications typically pair a single exposure with a single outcome. They show that momentary reports are sensitive to context, but they rarely compare several perceptual dimensions across the settings people occupy. Practices within the GEMA community have also become more standardized and reproducible in recent years, through shared system architectures and reporting guidelines (Gharani et al., 2021; Kingsbury et al., 2024). In population health research more broadly, integrating multiscale geospatial environmental data remains a methodological challenge (Cui et al., 2022). Wearable devices are increasingly used in related applications, such as collecting occupants' physiological data in indoor environmental quality studies (Abboushi et al., 2022), and smartwatches specifically have been found useful for measuring ambient noise (Fischer, Schraivogel, Caversaccio, & Wimmer, 2022). Crowdsourced deployments of related sensing methods have been framed as citizen science, with volunteers using smartphones to collect urban noise data (Zipf, Primack, & Rothendler, 2020). In these studies the device mainly records exposure or physiology, and the perception of the person carrying it is either not collected or elicited separately. Image-based surveys have the opposite limitation, eliciting perception without lived exposure (Ito et al., 2024; Quintana et al., 2025). Delivering the survey on the wearable itself addresses both limitations. A short question answered on the wrist can be repeated many times per day at low burden and paired with the device's sensor record of the same moment.

2.2. Heat and noise as everyday exposures across urban settings

In offices, noise from different sources affects perceived productivity and prompts a range of coping strategies (Appel-Meulenbroek, Steps, Wenmaekers, & Arentze, 2020). Individuals differ in how they perceive the same thermal conditions (Wang et al., 2018). This variation has motivated personal comfort models built for the individual instead of the building population (Arakawa Martins, Soebarto, & Williamson, 2022). Thermal sensitivity also varies with context, for example between air-conditioned and naturally ventilated buildings (Rupp, Parkinson, Kim, Toftum, & de Dear, 2022). The evidence on both exposures is, however, largely organized by setting. Large office occupant surveys identify acoustics, sound privacy, and temperature among the most common sources of dissatisfaction (Graham, Parkinson, & Schiavon, 2021; Parkinson, Schiavon, Kim, & Betti, 2023). Separate studies address distractions in the home-work environment (Bergefurt et al., 2023) and the perceived exposure of commuters on the move (Marquart, Schlink, & Nagendra, 2022). Because each setting is studied with its own instrument and population, it is difficult to compare how the same people experience heat and noise as they move between homes, offices, classrooms, transport, and outdoor spaces. Geolocated momentary reports can also be linked to digital-twin models of cities and pedestrian comfort (Ignatius et al., 2024). A consistent cross-setting characterization would give such models a useful baseline.

2.3. Prior work with the Cozie platform and this deployment

Open-source smartwatch platforms such as Cozie can interface with people in their natural settings, collecting micro-survey responses alongside the watch's onboard physiological and environmental sensor data (Tartarini, Frei, Schiavon, Chua, & Miller, 2023). They also support two-way communication, such as delivering just-in-time messages back to participants (Miller, Chua, Quintana, et al., 2025). These platforms have been used to characterize perceived conditions across indoor and outdoor urban settings (Tartarini et al., 2023) and in bedroom sleep environments (Miller, Chua, Frei, et al., 2025). The smartwatch micro-survey approach was first proposed as a framework for just-in-time adaptive interventions for building occupants (Miller, Chua, Frei, & Quintana, 2022). The Singapore deployment analyzed here has supported two further lines of work. Its data formed the basis of a public machine-learning competition that treated the responses as prediction targets, predicting smartwatch-reported heat and noise from digital-twin metrics (Miller, Ibrahim, et al., 2025; Miller et al., 2023). The same field protocol was also used to evaluate whether just-in-time messages nudge heat and noise mitigation behaviors, with a focus on intervention outcomes (Miller, Chua, Quintana, et al., 2025). Neither line characterizes the reported experience itself, describes the coverage and structure of the full dataset, or compares that experience across the everyday spaces in which it was recorded. The present paper provides that characterization. It documents the dataset's spatial, temporal, and participant coverage and compares thermal preference, noise, social context, and activity across six everyday space types using descriptive, effect-size-based methods. Together, these provide a descriptive baseline that complements the predictive and intervention studies.

3. Methods

This study characterizes human-experience data collected in Singapore using the Cozie platform, an open-source application for the Apple Watch that administers watch-based micro-surveys and records the wearer's onboard physiological and environmental data from the sensors (Tartarini et al., 2023). The overall approach was first proposed as a smartwatch-driven just-in-time intervention framework for building occupants (Miller et al., 2022). The deployment has since been described in a competition introduction and a competition overview and results paper (Miller, Ibrahim, et al., 2025; Miller et al., 2023), and the same platform underpins related work on just-in-time adaptive interventions (JITAI) for heat and noise (Miller, Chua, Quintana, et al., 2025). The design follows many of the established practices for environmental surveys and ecological momentary assessment and its geographically explicit extension, including recommendations for sampling, scheduling, and transparent reporting (Graham et al., 2021; Kingsbury et al., 2024; Liao, Skelton, Dunton, & Bruening, 2016; Reichert et al., 2020). Because the platform continuously records location and physiological signals, the deployment also considers the issues raised for GPS- and mobile-sensing studies (Fuller, Shareck, & Stanley, 2017; Kerr, Duncan, & Schipperijn, 2011). The remainder of this section describes the field study and participants, the structure of the micro-survey, the linked wearable, weather, and urban-context data, the processing steps that produce the survey-level dataset, the coverage and distributions of that dataset, and the analytical methods used to characterize the reported experience.

3.1. Crowdsourced watch-based micro-survey and physiological and environmental data

The field study methodology reported here is the same deployment described in detail by Miller, Chua, Quintana, et al. (2025), and the following is a summary of that protocol. The key elements of the protocol that are used in this analysis are illustrated in Figure 1. Participants were recruited through the National University of Singapore (NUS) Student Work Scheme and by word of mouth, producing a pool made up mostly of NUS students together with a smaller number of adults from outside the university. The cohort was predominantly young, with roughly 73% aged 18 to 30, and comprised 39% male and 61% female participants, and ethical approval was granted by the NUS Institutional Review Board. Each participant used an Apple Watch running the Cozie application paired with an iPhone, and loan devices were provided to those who lacked a compatible smartwatch. Participants were asked to wear the watch during working hours, at least from 9 AM to 7 PM on weekdays, for approximately four weeks. They were instructed to complete 100 micro-surveys together with weekly surveys and an in-person exit interview. Some participants provided responses outside the parameters in these guidelines, and those submitted within Singapore were retained in this analysis.

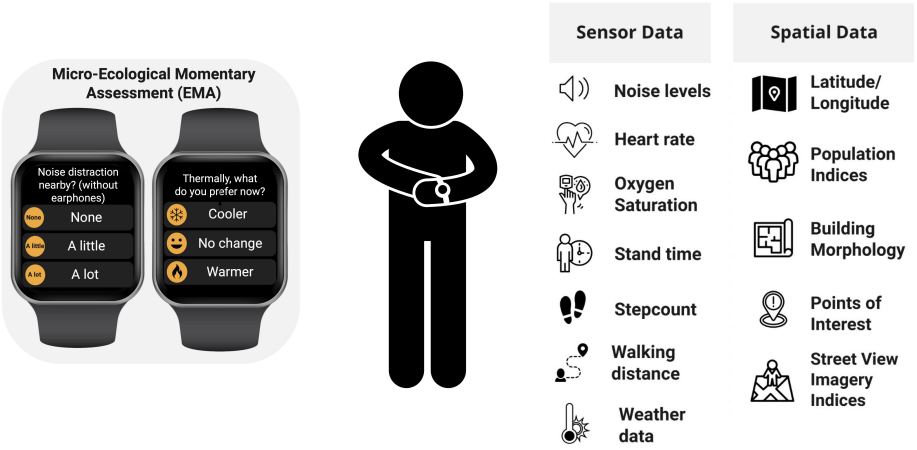


Figure 1. Overview of study design linking crowdsourced smartwatch micro-surveys (simple examples of questions on left) to the wearable sensor, weather, and urban spatial data each response is joined to (right) (Adapted from (Miller et al., 2022)).

As shown in Figure 1, each watch-based micro-survey response is time-stamped and geolocated, and is paired with the wearer's onboard physiological signals and with linked outdoor weather and urban-context data. The present analysis draws on a slightly different prepared dataset from this deployment, comprising 9,962 micro-survey responses from 106 participants, and focuses on characterizing the reported experience rather than the intervention outcomes.

Figure 2 illustrates the set of micro-survey questions answered directly on the watch face at each full submission instance. The questions are answered in a fixed order that the survey response flow later in this section follows column by column. A response first records the level of nearby noise distraction (*None*, *A little*, or *A lot*), and when noise is present, follow-up questions capture the kind of noise and whether earphones are being worn. It then records the participant's thermal preference (*Cooler*, *No change*, or *Warmer*) and the reported location or space type, with follow-up questions on the

kind of office or mode of transport when those spaces are selected. The survey closes by recording whether the participant is *Alone* or in a *Group* and the activity being performed, with the activity options depending on whether the participant is alone or with others. Several of these questions are conditional, so later items, such as noise kind, earphone use, and activity, are answered only for responses that reach them.

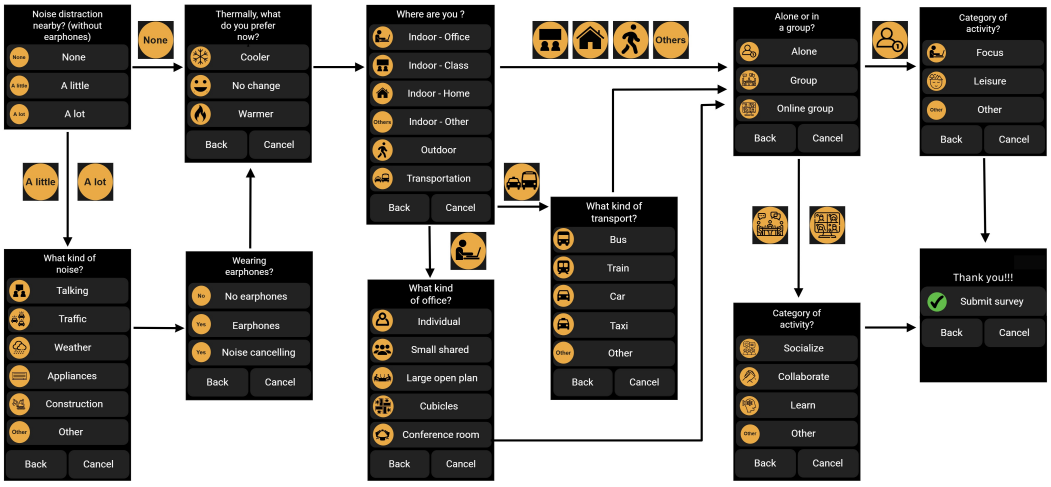


Figure 2. Full smartwatch micro-survey structure showing the left-to-right flow of watch screens from noise distraction, with its conditional noise-kind and earphone follow-ups, through thermal preference, location with its office and transport follow-ups, social context, and activity, to submission (Adapted from (Miller et al., 2022)).

Each geolocated response was linked to a set of urban-context indicators derived with Urbanity, an open-source tool that builds feature-rich urban networks from open geospatial data (Yap et al., 2023). These indicators summarize the built environment around each response location, including streetscape view indices for greenery, sky, buildings, and roads, as well as visual complexity, building count, population, and nearby points of interest across categories such as commercial, food, and recreational uses. These indicators are part of the linked dataset and supported the earlier prediction work on this deployment (Miller, Ibrahim, et al., 2025; Miller et al., 2023). This paper focuses on the self-reported experience and does not analyze them, but it documents them as covariates available for future work.

3.2. Data processing and linkage

The raw deployment data combines high-frequency watch and health streams with the sparser watch-survey responses in a single time-indexed record. Survey responses were first screened to those triggered on the watch that carried valid, non-zero coordinates falling inside the Singapore bounding box, which produced the 9,962-response analysis set. For each retained response, the seven temporal watch-health streams were aggregated over three pre-survey windows of six hours, one hour, and ten minutes, summarizing each window by its count, mean, spread, and extremes, and by an accumulated total for step count, walking distance, and stand time. The nearest weather-station observations were aggregated over the same three windows, and a heat index was derived from air temperature and relative humidity. These health and weather aggregates were then merged back onto the survey rows and checked for consistency, yielding

one analysis-ready record per response that includes the original survey and location fields, along with the linked physiological, weather, and urban-context features. For the spatial-coverage maps only, responses were further restricted to a frame around Singapore's main island (latitude 1.26 to 1.46°N, longitude 103.66 to 104.00°E). This excluded 31 responses with outlying GPS fixes and left 9,931 mapped responses, and all other analyses use the full 9,962-response set. The mapped responses are shown as a point-density surface, in which each response contributes within an 11-pixel radius of influence on the web-map tiles. They are also shown as counts aggregated by rounding each location to a 0.01° latitude-longitude grid, with cells of approximately 1.1 km at Singapore's latitude.

3.3. Dataset coverage and characterization

The analysis-ready dataset was characterized for its spatial, participant, and temporal coverage and for the distributions of the survey responses and linked measurements. These characteristics define the scope of the comparisons in Section 4.

3.3.1. Spatial, participant, and temporal coverage

Figure 3 shows the mapping of the responses across Singapore. The densest grid cells sit over residential estates and the NUS university campus, consistent with the roughly 45% of responses reported from home, while coverage thins quickly toward the periphery even as individual responses still reach the central business district and points along the transport network.

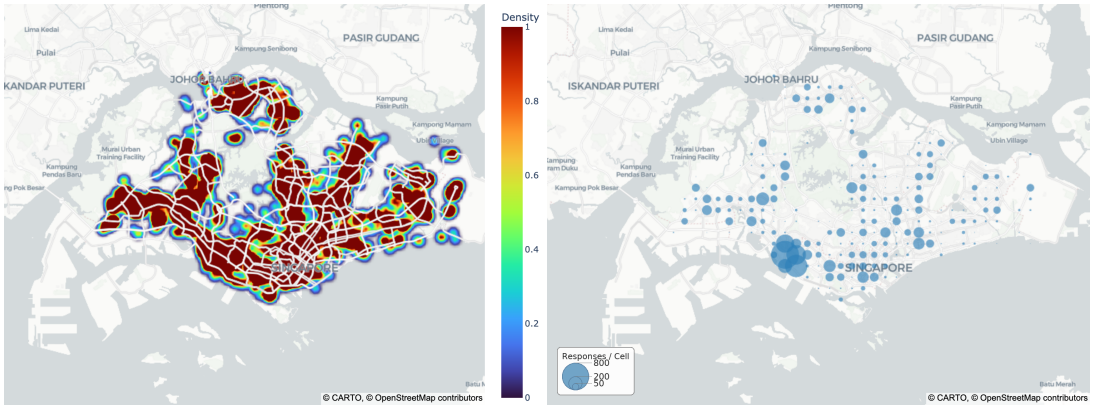


Figure 3. Spatial coverage of survey responses on Singapore's main island (N = 9,931 responses within latitude 1.26 to 1.46°N and longitude 103.66 to 104.00°E, excluding 31 outlying GPS fixes from the 9,962-response dataset), shown as a point-density surface with an 11-pixel radius of influence per response (left) and response counts aggregated into 0.01° latitude-longitude grid cells of approximately 1.1 km (276 cells, top cell 897 responses; right).

Figure 4 illustrates the breakdown of data collection across the cohort of 106 participants. Contribution is spread evenly enough that no small group dominates the dataset, with a median of 98 responses per person, close to the 100-survey target, and the most active tenth of participants supplying only about 12% of the total. The range from 17 to 158 responses does mean that some participants are somewhat unequally represented, but the analyses that follow rely on within-space proportions and effect sizes rather than raw response counts.

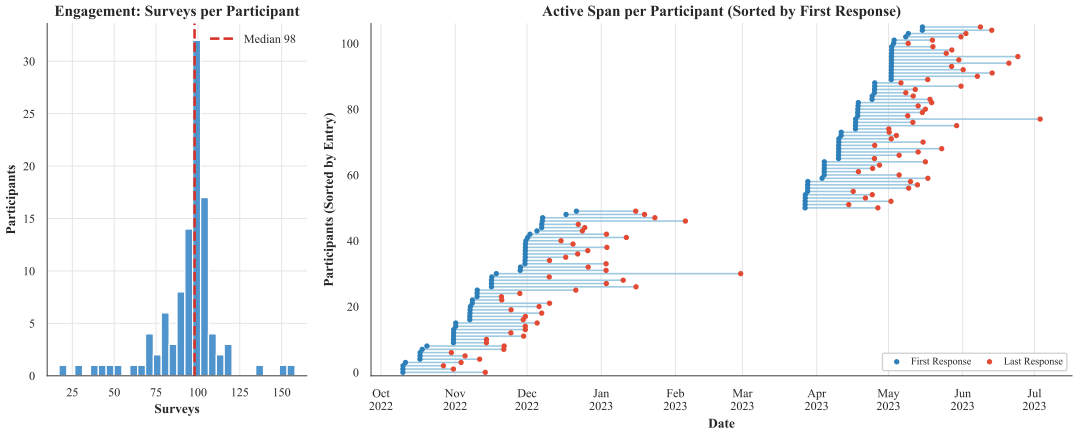


Figure 4. Participant engagement over the deployment: the distribution of surveys per participant (left; median 98) and each participant's active span from first to last response (right).

The temporal cadence in Figure 5 reflects a protocol built around the working day. Responses concentrate sharply between 9 AM and 7 PM and peak in the mid-afternoon, and weekdays account for all but about 17% of the data, so the dataset captures waking, active hours more than nights or weekends. The calendar-week panel shows the response volume rising and falling with the phased recruitment of participants.

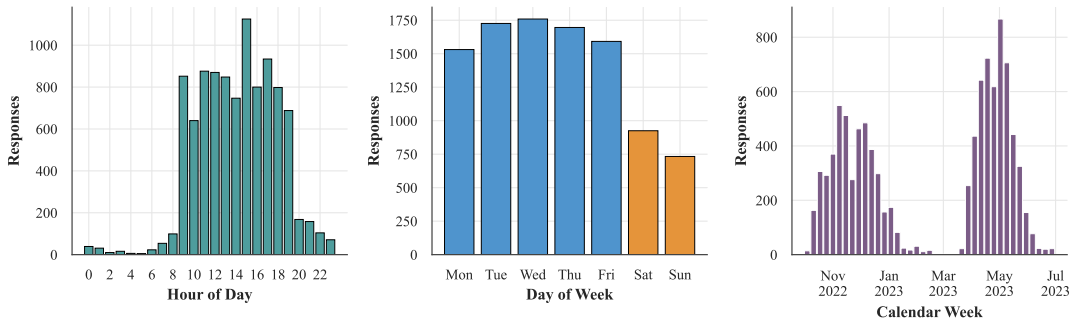


Figure 5. Time frames of survey responses in Singapore local time by hour of day (left), day of week (middle), and calendar week (right).

Timing is normalized within each space type in Figure 6, so it reads as a set of temporal signatures for when data is provided in that category. Offices and classrooms are almost exclusively weekday, daytime settings, with about 96% of office responses on weekdays, while home stretches into the evening and holds the largest weekend share of any space. The commute environments have a unique fingerprint, as outdoor and transportation responses have peakes in the morning and early evening following commute patterns.

3.3.2. Survey responses and linked data coverage

Figure 7 gives the overall distribution of every question and these data provide the breakdown of the participant self-assessments. Thermal preference is asymmetric, with a small majority wanting *No change* (54%), with much of the rest wanting to be *Cooler* (40%), and *Warmer* more rare (6%). This pattern is expected for a warm, humid

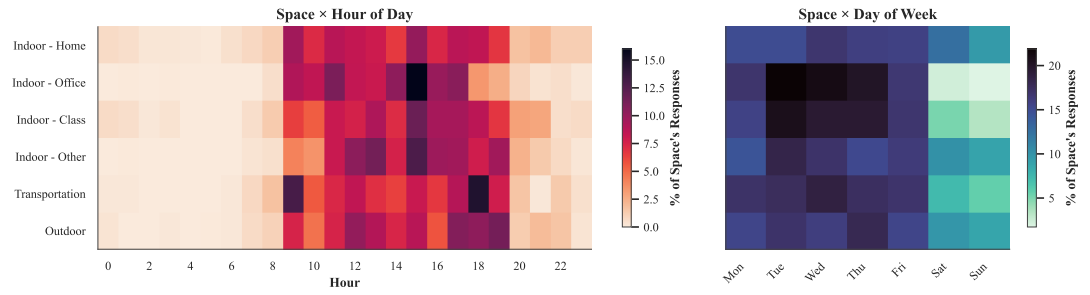


Figure 6. Row-normalized response timing by space type, showing each space's share of responses across hour of day (left) and day of week (right).

climate, though it also reflects that many indoor spaces are air-conditioned rather than exposed to the outdoor weather. Perceived noise distraction is more balanced, with roughly equal shares reporting *None* (46%) and *A little* (44%) and only about one in ten reporting *A lot*. Noticing some noise is the norm, while a strong distraction is uncommon. Where noise is present, the follow-up question most often attributes it to people talking (44% of noise-type responses), well ahead of appliances and traffic. Most responses come from home (45%), participants are usually *Alone* (72%) rather than in a *Group*, and the activity questions show focused work and leisure in solitary moments and socializing in group ones. Earphone (or audio headphone) use is uncommon, even though the question is included in about half of surveys, as most responses report *No earphones*, and active noise canceling is rare. Therefore, participants seem to tolerate rather than mask their sound environment. The coverage panel shows that linked analysis is feasible with the one-hour pre-survey window, including weather for about 96% of surveys and heart rate, movement, and measured noise for roughly 90%, so nearly every self-report has matching sensor data.

The same responses appear as a flow through the questions in Figure 8, which makes the conditional structure clearer through a Sankey diagram whose columns follow the question order of the micro-survey in Figure 2, each labeled with its survey question and color. Reading from left to right, each answer splits into the next question, the conditional branches, such as noise kind and earphones, carry only half of the responses that reach them, and activity is divided by whether the participant is *Alone* or in a *Group*. The ribbon widths are drawn to scale, so the proportion of every category shows that the common categories such as *No change* or *Cooler* are more prominent.

The linked wearable health streams are shown in Figure 9 with each parameter summarized over the six-hour, one-hour, and ten-minute windows preceding a survey. Median step count and walking distance fall as the window narrows, from about 85 to 42 steps per reading between the six-hour and ten-minute windows, because a short look-back captures only the last few minutes of activity. Heart rate is more stable at roughly 80 bpm across all three windows. Coverage moves the other way and limits how tight the window can be. The six-hour window carries movement and noise for over 95% of surveys, but at ten minutes, audio coverage falls to about a third, and step coverage to little more than half. Resting heart rate and blood-oxygen streams are sparse in every window. The one-hour window is used as the aggregate method throughout the rest of this analysis, since it maintains coverage near 90% while staying close to the survey moment.

Figure 10 summarizes the nearest-station outdoor weather over the same windows, and its distributions are narrow and stable, as expected for an equatorial climate. Air

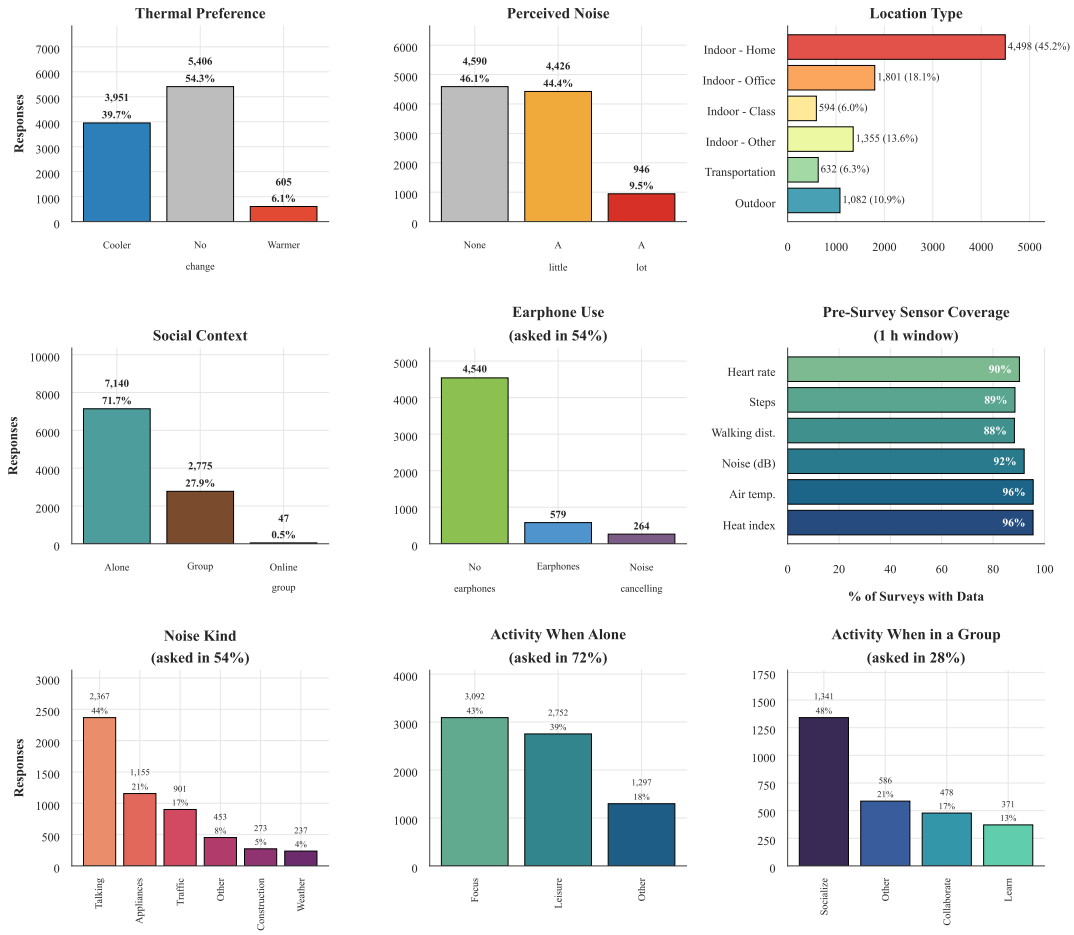


Figure 7. Response distributions for each survey item and pre-survey linked-sensor coverage (N = 9,962 responses, 106 participants).

temperature has a median near 29 °C and relative humidity near 73%, which gives a heat index around 34 °C that changes little across the windows. The outdoor thermal environment is consistently warm, whatever the window length. Rainfall is the one strongly window-dependent feature because it accumulates over time, with about 28% of six-hour windows containing some rain against only 5% of ten-minute windows. These values describe outdoor conditions at a station, a median of roughly 3 km, from the response. It should be noted that these data are not the participant's actual microclimate, which is often indoor and air-conditioned.

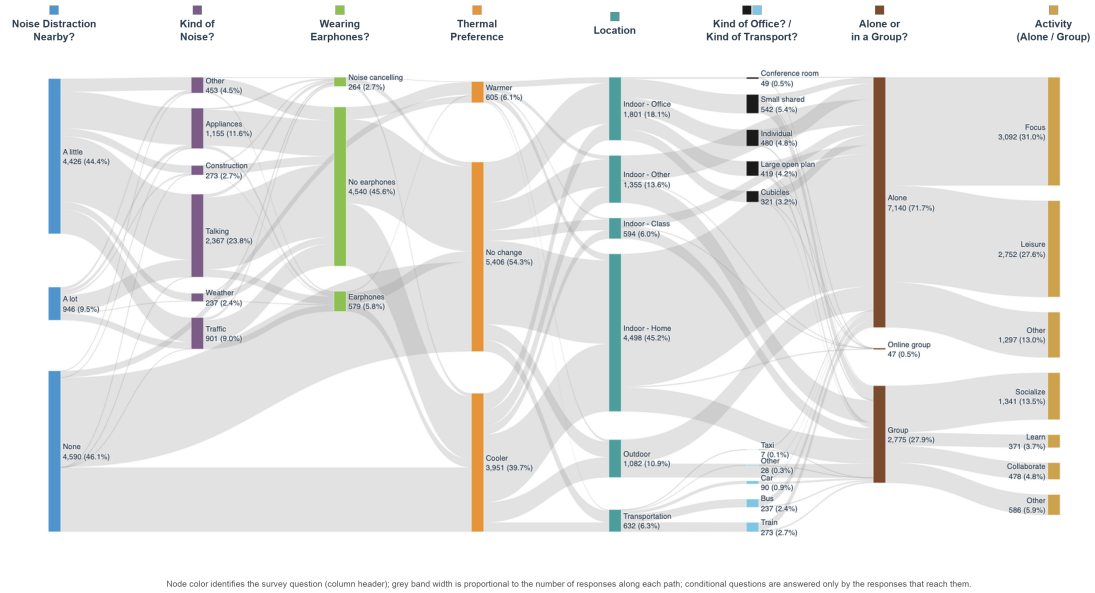


Figure 8. Sankey diagram of the conditional response flow through the micro-survey, with one column per survey question (labeled at the top, with the color swatch matching that question's nodes), office and transport type sharing the sixth column, the alone and group activity options sharing the last column, and grey band width proportional to the number of responses along each path.

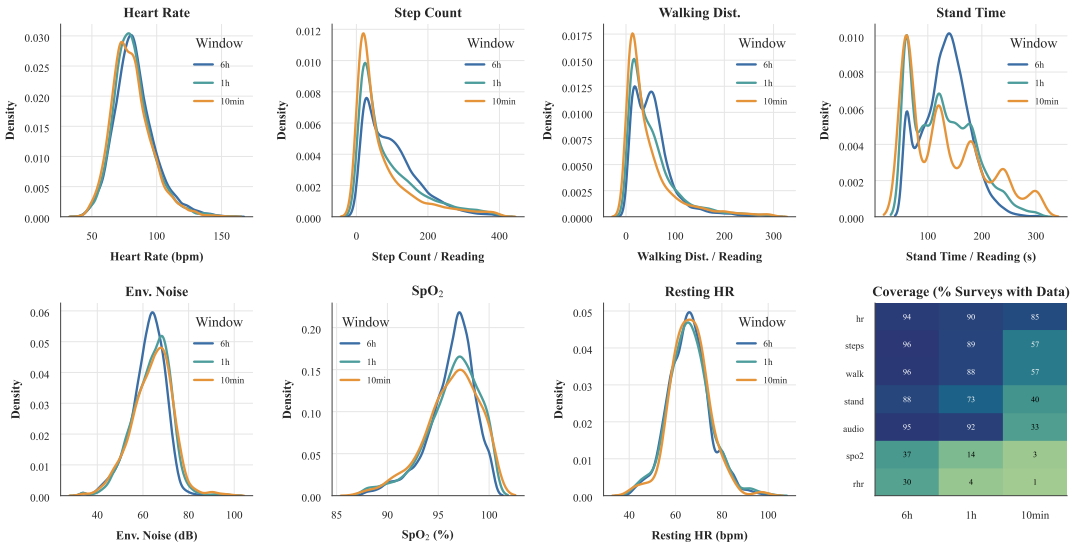


Figure 9. Distributions of pre-survey wearable-health metrics across the 10-minute, 1-hour, and 6-hour aggregation windows: seven wearable health and audio streams as overlaid per-window density curves, plus a metric-by-window data-coverage heatmap.

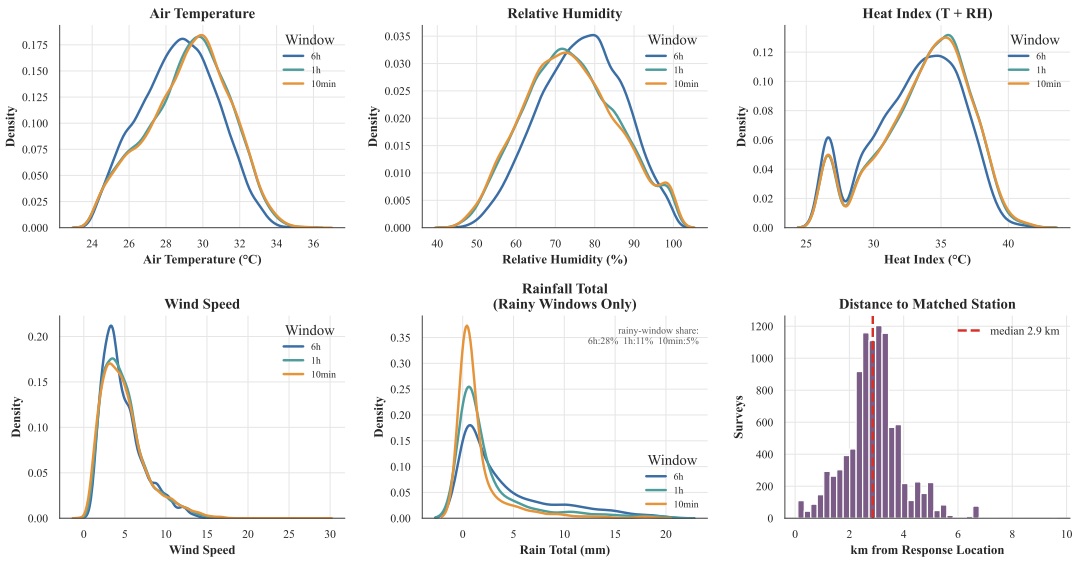


Figure 10. Nearest-station outdoor weather across the 10-minute, 1-hour, and 6-hour pre-survey windows, with distributions for five ambient metrics and the distance to the matched station (median 2.9 km).

3.4. Perceptual characterization methods

The reported experience was then characterized against the reported space type using the analysis-ready record described above. Each response was categorized into one of the space types based on that survey response, and each categorical perception variable was cross-tabulated with the space type to form the contingency tables used below. Conditional questions, such as noise type and earphone use, have lower coverage, so their profiles describe only the responses for which follow-up was observed. Because responses are clustered within participants, the significance markers reported below are treated as descriptive screens, and effect sizes and the coherence of contrasts across related variables are used as the primary read-outs. The three methods below produce the perceptual characterization used for analysis.

3.4.1. Conditional perception fingerprints

For each space type s and response category r, the conditional proportion is the maximum-likelihood estimate of the categorical distribution within that space,

$$\hat{P}(\text{response} = r \mid \text{space} = s) = \frac{n_{rs}}{n_s}, \tag{1}$$

where n_{rs} is the number of responses of category r in space s and n_s is the total number of responses in that space. Each row, therefore, describes a single space's response profile and sums to 1 within each survey variable. No smoothing or shrinkage is applied, so cells based on few responses are noisier than their shading suggests.

3.4.2. Relative distinctiveness through lift

To separate what is distinctive to a space from what is simply common everywhere, each conditional proportion is divided by the corresponding corpus-wide proportion

to give the lift,

$$\text{lift}(r, s) = \frac{P(\text{response} = r \mid \text{space} = s)}{P(\text{response} = r)}. \tag{2}$$

A value of one indicates a response that is as common in the space as in the corpus overall, values above one indicate over-representation, and values below one indicate under-representation.

3.4.3. Association strength and cell-level contrasts

The overall association between space type and each perception was tested with a Pearson chi-square statistic on the $r \times k$ contingency table,

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \tag{3}$$

where O_{ij} and E_{ij} are the observed and expected counts under independence. Because χ^2 grows with sample size, it was rescaled to a bounded effect size using Cramér's V,

$$V = \sqrt{\frac{\chi^2}{n\,(\min(r, k) - 1)}}, \tag{4}$$

interpreted with conventional descriptive bands (negligible < 0.1, weak 0.1–0.2, moderate 0.2–0.3, strong > 0.3). To identify which individual space and response combinations drive the association, adjusted standardized residuals were computed for each cell,

$$z_{ij} = \frac{O_{ij} - E_{ij}}{\sqrt{E_{ij}\,(1 - r_i/n)\,(1 - c_j/n)}}, \tag{5}$$

where r_i and c_j are the row and column totals and n is the grand total. Under independence, each z_{ij} is asymptotically standard normal, so cells with $|z| > 1.96$, 2.58, and 3.29 exceed the conventional 0.05, 0.01, and 0.001 thresholds and mark the space and response pairs shown as over- or under-represented.

4. Results

The results address the two research questions in turn. Section 4.1 relates thermal preference and noise distraction to the linked outdoor heat and measured sound level and examines differences between participants. Section 4.2 characterizes how the reported experience differs across the six everyday space types.

4.1. Thermal preference and noise in relation to linked measurements

The two dominant strands of the reported experience, wanting to be cooler and being distracted by noise, were examined before they were related to space type. Cooling preference relative to outdoor heat, across participants, is shown in Figure 11, and the

demand is real but only loosely tied to the outdoor thermometer. Across all responses, the share of *Cooler* answers rises with outdoor heat, from about a third when the pre-survey heat index is near 30 °C to roughly half above 37 °C. This situation is expected for a tropical setting, though the outdoor reading is only a proxy for the participant's actual microclimate, which is often a controlled indoor space rather than the outdoor weather. The link between people is weaker than within the heat range. A participant's mean outdoor temperature explains little about who wants cooling ($r \approx 0.33$), and the per-participant panel is very wide, with the *Cooler* share ranging from under 10% for the most heat-tolerant participants to nearly 80% for the most cooling-seeking participants. This spread is a key takeaway of the figure, showing that cooling preference is as much a personal trait as a response to measured conditions, the individual variation that person-level sensing is designed to capture.

The same type of characterization for the sound environment is given in Figure 12, using the linked one-hour sound-level measurement, and here the self-reports and the sensor agree well. Measured sound level rises steadily with perceived noise, from a median near 62 dB for *None* to 66 dB for *A little* and 70 dB for *A lot*. The separation is consistent despite the coarse three-point scale. The measured levels also track where people are, with home and office the quietest settings at around 62 dB while other-indoor, transportation, and outdoor environments sit near 69 to 70 dB, and the sources most often reported as loud, traffic and talking, carry the highest levels. When analyzed together, the two figures show that thermal preference is a personal signal that only loosely tracks the surrounding environment, while noise perception is tied to a measurable exposure that varies with the space around the participant.

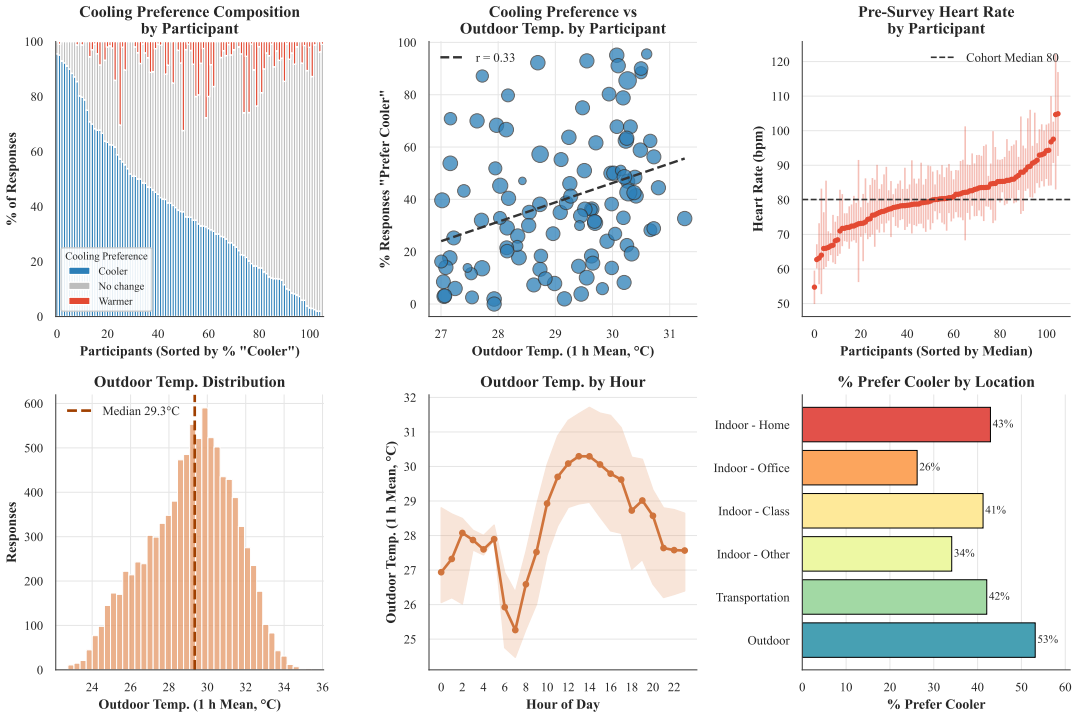


Figure 11. Participant-level and contextual patterns in cooling preference, relating per-participant composition, outdoor temperature, heart rate, and location type.

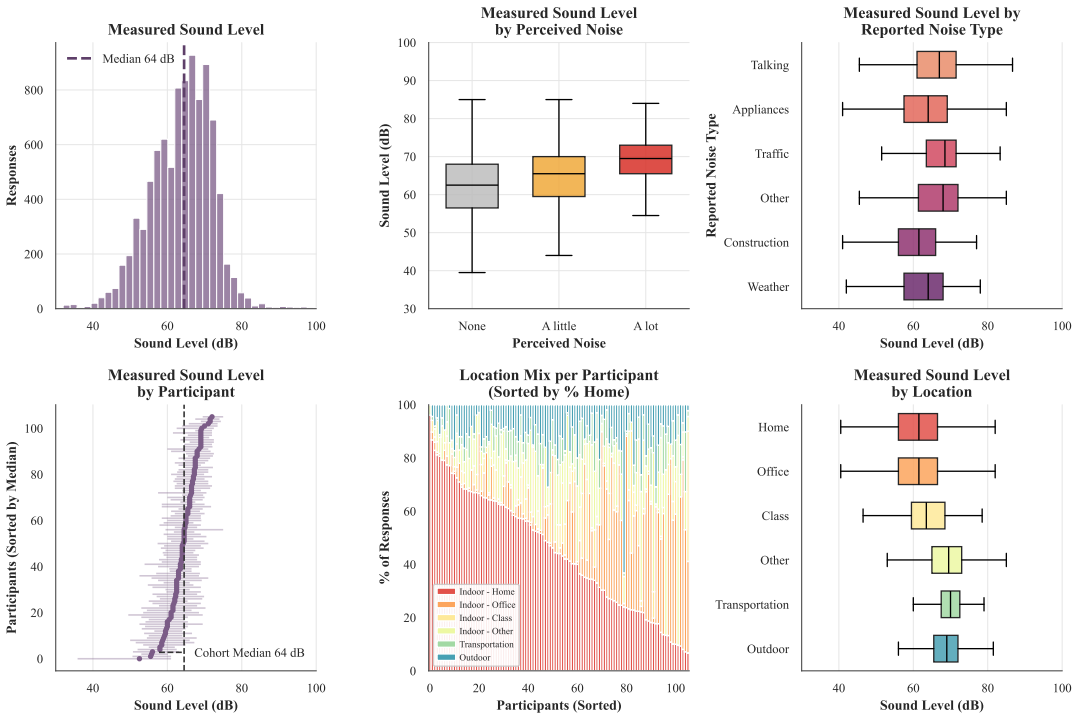


Figure 12. Measured sound level (1-hour pre-survey mean) summarized across subjective, participant, and location contexts.

4.2. Perceptual fingerprints and association with space type

With each perception characterized on its own, the analysis now asks how the reported experience differs across the spaces in which it was recorded. It does so in three layers that build on one another. The first layer, in Figure 13, reads each space type as a profile by normalizing within its own row. Every cell gives the probability of a response given the space, so each row describes the mix of experiences in that setting rather than how many responses it contributed. Read this way, the spaces separate into recognizable characters. Home is a quiet, solitary setting, with most responses reporting *None* for noise and about 84% reporting *Alone*, while classrooms and other-indoor spaces are mostly social. The commute spaces stand out for discomfort, with the highest shares of *A lot* of noise, reaching roughly one in four responses in transportation against only about 3% at home. Thermal preference shifts more gently, from only about a quarter of office responses wanting *Cooler* to just over half of outdoor responses. This fits the heavily air-conditioned office and the exposed outdoor setting, and it is a reminder that thermal preference reflects the immediate controlled environment more than the outdoor weather. The fingerprint is intuitive but has one limit, since some responses are common everywhere and a high within-space probability does not by itself mean the response is characteristic of that space.

Figure 14 shows the second layer that divides each within-space probability by the response's overall proportion in the dataset. A lift above one marks a response that is more common in a space than in the data as a whole, and a lift below one marks one that is rarer. This turns a profile into a signature, because it isolates what is distinctive to a space rather than what is simply frequent. The contrast sharpens the same story and adds detail the fingerprint hides. *A lot* of noise is over-represented by about 2.6

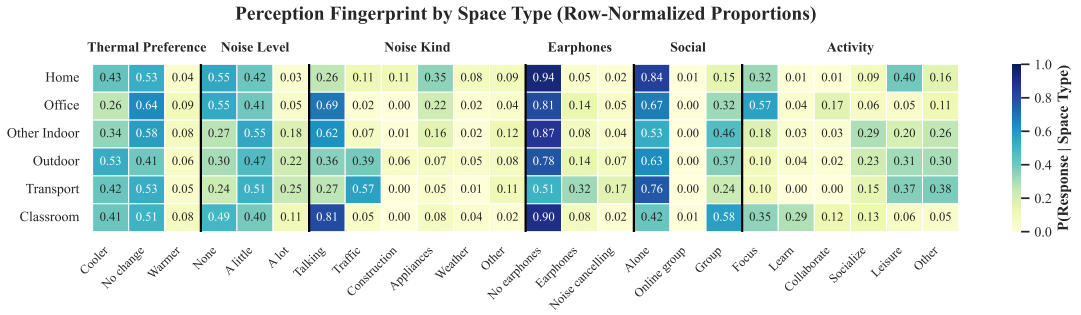


Figure 13. Conditional perception fingerprint by space type in a heatmap of $P(\text{response} \mid \text{space type})$ with space types in rows and response options (grouped by survey dimension) in columns.

times in transportation and 2.4 times outdoors, *Group* is over-represented most in classrooms (lift near 2.1), and *Warmer*, a rare response overall, is concentrated in the office (lift near 1.6) where air-conditioning can overshoot. Lift also makes the source of noise a spatial fingerprint in its own right, as traffic accounts for most reported noise in transportation and a large share outdoors, while talking dominates the noise reported in classrooms and offices. The limit of lift mirrors that of the fingerprint, since it flags distinctive combinations but does not show whether any single contrast is larger than chance would produce.

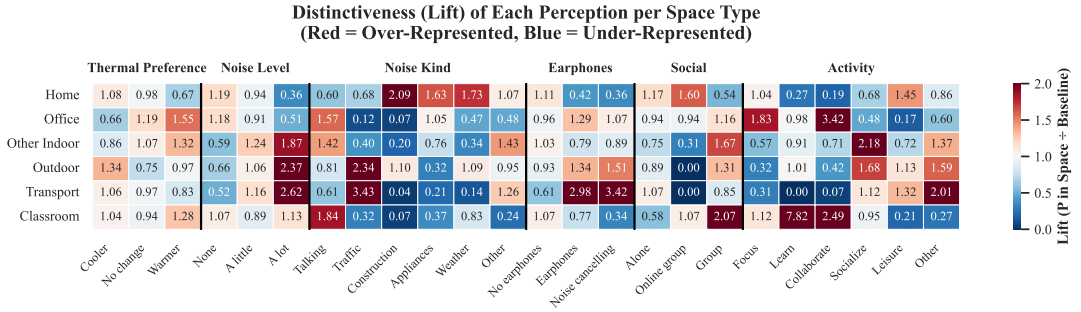


Figure 14. Relative distinctiveness of perceptions by space type, shown as lift, $P(\text{response} \mid \text{space type})/P(\text{response})$, with space types in rows and response options (grouped by survey dimension) in columns.

Figure 15 illustrates the third layer of analysis that adds the statistical dimension the first two lack. It compares each space-response combination against what independence would predict, using adjusted standardized residuals whose sign indicates the direction of the effect and whose magnitude is standardized to a standard-normal scale. This layer confirms which distinctive contrasts are large relative to sampling noise, and it agrees with the overall effect sizes. The association with space type is strongest for what a person is doing and hearing, with Cramér's V near 0.38 for group activity, 0.35 for solitary activity, and 0.28 for the kind of noise, and weakest for thermal preference at about 0.12. The residuals make the behavioral picture explicit, as transportation and outdoor spaces stand out for loud, traffic-driven noise, classrooms and offices for talking and for group or focused activity, and home for quiet, solitary responses. Thermal preference produces only muted residuals scattered across spaces. Because the responses are clustered within 106 participants, the significance markers are read as descriptive screens rather than formal tests, and the agreement of all three layers on

the same ordering is the stronger evidence. The lesson across the three layers is that space type is written far more clearly into what people are doing and hearing than into how warm they feel. This weak thermal signal is consistent with people often being in controlled indoor settings rather than being exposed to outdoor weather.

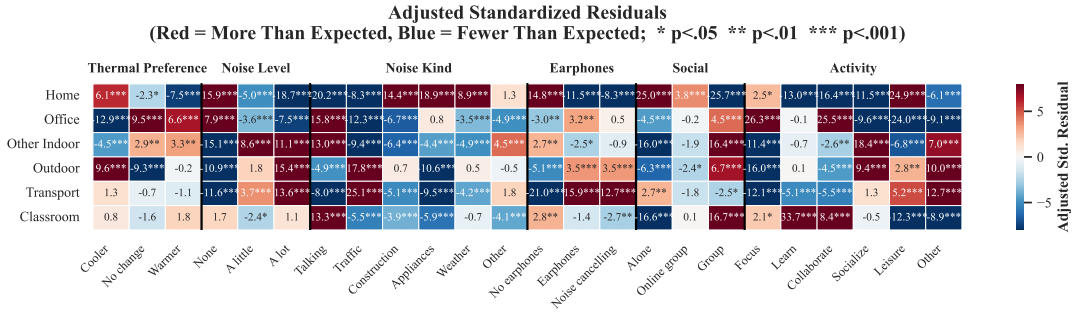


Figure 15. Adjusted standardized residuals for space-type by response combinations, where red cells occur more often and blue cells less often than expected under independence.

5. Discussion

The three analytical layers converge on a coherent characterization of how urban space types differ in reported experience. The clearest separations are behavioral and acoustic rather than thermal, since what people are doing and hearing identifies a space more sharply than how warm they feel. Tables 1 and 2 collect the findings that reinforce prior expectation and establish this baseline character of each setting, grouped by the acoustic environment and by social context and thermal comfort. Each entry pairs a short insight with the supporting detail drawn from the fingerprint, lift, and residual analyses of Section 4.2.

Table 1. Space-type findings that reinforce prior intuition for the acoustic environment.

Commute spaces are the noisiest	Transportation and outdoor responses carry the highest *a lot* noise shares (≈ 25% and 22%), against only about 3% at home.
Home and offices are the quietest settings	Both sit near a 62 dB median sound level, the lowest among the measured spaces.
Perceived loudness tracks the setting	Louder spaces attract more *a lot* noise reports, so subjective loudness follows the physical environment.
Traffic defines commute noise	Traffic is the dominant reported noise source in transportation (≈ 57%) and outdoors (≈ 39%).
Talking defines indoor noise	Talking dominates the acoustic complaints of classrooms (≈ 81%) and offices (≈ 69%).

The findings show that a watch-based micro-survey recovers the structure of human perception in the urban context, some of which aligns with previous studies. Commute spaces carry the highest *a lot* noise shares, and traffic dominates their reported noise source. These results agree with mobile-exposure studies of commuters, which found the perception of noise and air pollution to be context-dependent and pairing mobile measurements with walking interviews to be advantageous (Marquart et al., 2022). They are also consistent with urban assessments that find core urban areas noisier than

Table 2. Space-type findings that reinforce prior intuition focused on social context and thermal comfort.

Home is overwhelmingly solitary	About 84% of home responses report being *alone*, the most solitary setting measured.
Classrooms are the most social setting	Classrooms are the most group-oriented space, with a group lift near 2.1.
Activity is the strongest spatial marker	Group activity is the response most tightly associated with space type (Cramér's V ≈ 0.38).
Offices show the lowest cooling demand	Only about 26% of office responses want *cooler*, consistent with heavy air-conditioning.
Outdoor spaces show the highest cooling demand	Just over half ($\approx 53\%$) of outdoor responses want *cooler*.

peripheral areas (Kumar & Pandey, 2013), and with survey evidence that annoyance from residential road-traffic noise depends on non-acoustic factors such as windows and household income as well as on exposure (Preisendörfer, Liebe, Bruderer Enzler, & Diekmann, 2022). That a wrist-based micro-survey, interviews, and remote sensing produce the same ordering of spaces gives more confidence than any single method would, and it supports treating the less obvious findings that follow as reliable.

However, beyond the intuitive elements, several results are less obvious and reshape how the characterization could be read. Thermal preference, the signal one might expect a tropical climate to dominate, is in fact the weakest spatial marker, while distinctiveness that raw proportions understate emerges only once responses are weighed against their overall frequency. Tables 3 and 4 gather these counterintuitive and analytical findings, separating the environmental and thermal surprises from the behavioral signatures and from the framing that effect sizes, rather than p-values, provide under participant nesting.

Table 3. Less intuitive environmental and thermal insights for further exploration.

Thermal preference is the weakest spatial signal	Its association with space type (Cramér's V ≈ 0.12) sits far below activity or noise.
Cooling demand separates spaces only modestly	Outdoor demand ($\approx 53\%$) exceeds home ($\approx 43\%$) by less than an air-conditioning-driven intuition would suggest.
Offices can over-cool	Wanting *warmer*, rare overall, concentrates in offices (lift ≈ 1.6), a signature of air-conditioning overshoot.
The kind of noise matters more than the amount	The source of noise fingerprints a space more sharply than how loud it is.
Lift exposes hidden distinctiveness	*A lot* of noise is $\approx 2.6\times$ over-represented in transportation relative to the whole corpus, a contrast raw proportions understate.

That activity is the sharpest spatial marker is consistent with behavioral theories of how occupants interact with building systems (Heydarian et al., 2020) and with evidence that movement and exposure are spatially structured within a single setting (Pollard et al., 2022).

5.1. Implications for sustainable urban design

These results help close the loop between how planners assume occupants behave and perceive their surroundings and how occupants actually report experiencing them. The analysis is descriptive and compares broad, self-reported space types. The im-

Table 4. Method-based and behavioral insights for further exploration.

Behavior separates spaces best	What a person is doing distinguishes space types more sharply than any environmental variable.
Home is distinctively, not just frequently, solitary	Even against an already alone-dominated corpus, home's alone lift exceeds one.
Commuting is loud and lonely	Transportation pairs loud, traffic-driven noise with high solitude ($\approx$ 76% alone), combining discomfort with isolation.
Three methods converge	Fingerprint, lift, and residuals agree on the same behavioral ordering, strengthening confidence beyond any single test.
Effect sizes, not p-values, are the valid read-out	Responses are nested within 106 participants, so significance markers act as descriptive screens rather than formal tests.

plications below therefore indicate where to target post-occupancy evaluation and follow-up measurement, and they are not quantified design prescriptions. Much of the picture reinforces intuition. The contribution lies less in any single finding than in quantifying and localizing these patterns from reports made on site, instead of recalled later or judged from images. The same instrument can also flag settings that perform better or worse than expected. This matters because how residents read an environment can affect outcomes such as self-rated health more than the measured environment itself (Y. Kim, 2016), so a design process tuned only to physical metrics can miss the perceptions that shape how a space is used. Large office-survey databases identify acoustics and temperature among the most common sources of occupant dissatisfaction (Graham et al., 2021; Parkinson et al., 2023), and noise among the main influences on workspace satisfaction (Frontczak et al., 2012). Localizing where these complaints arise is therefore directly useful for post-occupancy evaluation. Delivered on a consumer smartwatch, the micro-survey extends the mobile-crowdsourcing approach of collecting geolocated resident feedback (Phuttharak & Loke, 2020) into a continuous channel for post-occupancy evaluation and participatory planning, aligning with the broader agenda of human-building interaction for intelligent built environments (Becerik-Gerber et al., 2022).

From the acoustic perspective, the findings indicate where design attention is most likely to matter. Commute and outdoor spaces hold the loudest and most traffic-driven noise, with about a quarter of transportation responses reporting *a lot* of noise against only about 3% at home. Transit corridors and commute routes are therefore candidates for evaluating traffic calming, acoustic buffering, and quieter routing where distraction is worst. Indoors, talking dominates the noise reported in classrooms and offices more than traffic does. This echoes evidence that background speech is among the most disruptive office noise sources (Jahncke, Björkeholm, Marsh, Odelius, & Sörqvist, 2016) and the trade-off between privacy and communication in open-plan layouts (J. Kim & de Dear, 2013). Acoustic treatment and layouts that manage speech and reverberation are therefore the more relevant levers indoors.

The thermal results are related to energy policy and design considerations for a tropical, heavily air-conditioned city. Offices show the lowest cooling demand of any space, with only about a quarter of responses wanting *cooler* and occasional *warmer* requests where cooling overshoots. This is consistent with the possibility that relaxing indoor setpoints could save cooling energy at little cost to comfort, although testing that hypothesis would require measured indoor conditions. Outdoor spaces show the highest cooling demand, so shading, greenery, and other passive cooling of public spaces deserve priority when evaluating thermal relief. Quantifying their effect would require

microclimate measurement beyond this dataset. Thermal preference is the weakest spatial marker and reflects the immediate controlled setting more than the outdoor weather, so comfort here is set mainly by the indoor environments people occupy rather than by climate alone.

The layered reading used here fits a broader move toward multi-method assessment of urban perception that combines subjective, physiological, and spatial data, as in work that joins participatory mapping, neurophysiological response, and self-reported emotion to study perceived walkability (Yıldırım, Heyik, Güleç Ozer, & Sungur, 2026). Because the same watch-based platform has already supported city-scale prediction of noise distraction and thermal preference (Miller, Ibrahim, et al., 2025), the structure described here could feed predictive, just-in-time design tools embedded in urban digital twins (Lei et al., 2026) and the urban-scale occupant-behavior models used to anticipate building energy demand (Salim et al., 2020), tools that anticipate where and when discomfort arises. Heat and noise are only two of the dimensions such an instrument can capture, and the same loop could be closed for others.

5.2. Limitations

Generalizability is a limitation in this analysis as the findings come from 106 participants in a young and homogeneous cohort, with roughly 73% aged 18 to 30 and most recruited as university students. Recruitment was by convenience rather than through a representative sampling frame, so the reported patterns describe this cohort in Singapore and should not be read as population-level estimates. The setting itself also limits transfer, since Singapore is a dense, tropical, heavily air-conditioned city, and the thermal results in particular reflect that context more than a general climate response.

The structure of the data, rather than a controlled design, also shapes what the analysis can claim. Although there are 9,962 responses, they are nested within only 106 participants, so responses from the same person are not independent. We therefore treat the associations and significance markers as descriptive screens rather than formal tests, and we do not fit a model that accounts for this clustering. Perceptions are self-reported through single-item micro-surveys answered in the moment, and the reported space type is a participant label rather than a verified location, so both carry the usual subjectivity and recall effects of momentary self-report. Responses are also clustered in space and time, because participants report repeatedly from the same homes, workplaces, and routes and at similar times of day. Like participant nesting, this spatial and temporal autocorrelation reduces the effective sample size below the response count. It is a further reason to read the associations as descriptive.

Selective responding is a further source of bias, because completing each micro-survey depended on the participant choosing to respond. Moments when answering is inconvenient or unsafe, such as driving, active meetings, or intense heat or noise, are therefore likely to be underrepresented, while comfortable, static indoor states may be overrepresented. As a result, the space-type profiles describe the moments participants chose to report, not exposure-weighted time in each setting. The most extreme conditions are, if anything, likely to be understated.

Uneven coverage across the dataset qualifies the finer comparisons, beginning with the conditional follow-up questions such as noise kind, activity, and earphone use, which are answered only by the responses that reach them and so rest on smaller samples than the always-asked items. Within-space cells built from a few responses

are noisier than their shading suggests, because no smoothing or shrinkage is applied. Linked streams are also uneven, as some physiological signals, such as blood oxygen and resting heart rate, are sparse, and the early-morning hours are thinly sampled, with under 2% of responses before 08:00, so temporal patterns in those hours are weak. The space types themselves are unbalanced, with nearly half of all responses coming from home, so the corpus-wide baselines used for lift are weighted toward the most common settings.

Exposure and scope raise a final pair of limitations, beginning with the weather values, which are aggregated from the nearest station and describe ambient outdoor conditions rather than the microclimate or skin-level exposure each participant actually experienced, a coarse proxy especially for the indoor and air-conditioned settings where many responses were made. Finally, the deployment ran in two phases that differed in how just-in-time messages were triggered, and the present analysis pools across them, so it characterizes reported experience and does not isolate any effect the intervention messages may have had on what participants reported or did.

5.3. Future work

The most direct next step is to broaden the cohort through larger deployments that run fully on participants' own smartwatches with remote self-onboarding, which could reach thousands or tens of thousands of people and a wider mix of ages, occupations, and living situations. A more diverse cohort would allow the reported patterns to be tested for generalizability rather than described for one group, and replication in other cities would show which findings are specific to a dense, tropical, air-conditioned setting and which hold more broadly.

Future analyses could also treat the nested structure of the data directly, using mixed-effects or otherwise participant-grouped models that account for repeated responses from the same person, turning the descriptive associations reported here into estimates with proper uncertainty and supporting participant-grouped validation for any predictive work built on the same data. Exposure measurement could improve by pairing the micro-surveys with personal or wearable microclimate sensing, such as on-body temperature, humidity, and sound level, and with low-cost indoor air-quality sensors, on which researchers and practitioners have recently offered guidance (Parkinson et al., 2026). Such sensing would replace the coarse nearest-station weather with the conditions each participant actually experienced. This would matter most in the indoor and air-conditioned settings that dominate the responses. With such measured exposures and the urban-context indicators already linked to each response, future work could test whether environmental conditions shape comfort directly or through the perceived thermal and acoustic environment. It could also quantify how specific built-environment features, such as greenery, sky exposure, and building density, modify the experience within and across the space types used here.

The survey design itself could become more targeted, with prompts that adapt to where a person is and to the specific subjective question a deployment is meant to answer, concentrating responses where they are most informative and filling the thinly sampled settings and hours noted above. Building on this, a deeper characterization of participants who share behavioral patterns could establish behavioral archetypes, helping identify which patterns to encourage through design and which to discourage before they become common. In addition, longer and more structured deployments would extend the scope, since running over months rather than weeks would capture

seasonal and longer-term variation, and designs that separate or randomize the just-in-time intervention phases would let future work isolate the effect of the messages on reported experience and behavior, which the present pooled analysis cannot.

6. Conclusion

This paper characterizes the everyday human experience of urban space types in a dense tropical city using scalable, crowdsourced smartwatch micro-surveys. A field deployment of the open-source Cozie platform in Singapore collected 9,962 geolocated responses from 106 participants, each linking self-reported thermal preference, noise, social context, and activity to onboard physiological and linked weather data. The reported experience was characterized across six everyday space types through conditional perception fingerprints, relative distinctiveness, and effect-size-based association testing. The results outline that what people do and hear separates urban spaces more clearly than how warm they feel. Activity and the kind of noise are the strongest spatial markers, while thermal preference is the weakest, reflecting the heavily air-conditioned indoor settings that dominate daily life more than the outdoor climate. Commute spaces emerge as the loudest and most traffic-driven, indoor spaces are defined by talking, and cooling demand is lowest in offices and highest outdoors. Much of this reinforces intuition, but the value of the approach is that it recovers and quantifies these signatures directly from lived, in-situ reports rather than from assumptions built into design. The method also has clear limits, including a young and homogeneous cohort, selective responding, clustering within participants, places, and times, and ambient instead of personal exposure. Each of these limits points to a direction for future work. Taken together, the results show that a wrist-based micro-survey can provide a descriptive, human-centered baseline for targeting noise mitigation and energy-conscious, thermal-comfort-aware urban design, and they set the stage for larger and more diverse deployments that link the same reports to measured exposures and specific design features.

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CRediT authorship contribution statement

Clayton Miller: Conceptualization, Methodology, Software, Formal analysis, Resources, Writing – original draft, Visualization, Supervision, Project administration, Funding acquisition. Mario Frei: Conceptualization, Methodology, Software, Validation, Investigation, Data curation, Writing – review & editing, Project administration. Yun Xuan Chua: Conceptualization, Methodology, Validation, Investigation, Writing – review & editing, Project administration.

Declaration of generative AI use

During the preparation of this work, the authors used Runcell AI (Version 0.1.25), an AI coding and writing assistant built on the GPT-5.6 Terra (OpenAI) and Claude Opus 4.8 (Anthropic) large language models, to help organize and format the LaTeX manuscript, arrange figure and table placement, and copy-edit text written by the authors. The tools were not used to generate or analyze the study data, nor to create or alter any figure, its underlying values, or the reported results. All AI-assisted output was reviewed, edited, and verified by the authors, who take full responsibility for the originality, accuracy, integrity, citations, and final content of this manuscript.

Disclosure statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability statement

The data, code, and documentation supporting the findings of this study are available on GitHub at `https://github.com/buds-lab/cozie-singapore-journal`.

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